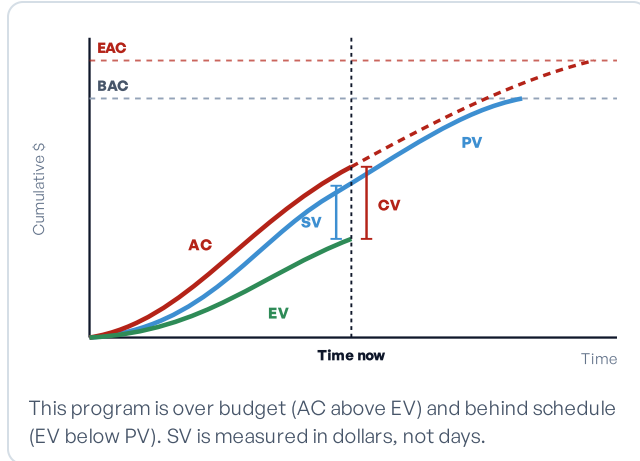


EVM Formulas Quick Reference

The numbers behind every program review, and how to read them.

1 The three numbers everything comes from



PV **Planned value.** Budget for the work scheduled to be done by now.
Also BCWS, budgeted cost of work scheduled

EV **Earned value.** Budget for the work actually done by now. The only number that measures progress.
Also BCWP, budgeted cost of work performed

AC **Actual cost.** What that work actually cost.
Also ACWP, actual cost of work performed

BAC **Budget at completion.** The whole planned budget: PV at the finish line.

2 Variances and indices

METRIC	FORMULA	GOOD	BAD	SAY IT OUT LOUD
Cost variance (CV)	$EV - AC$	+ under budget	- over budget	How much the work done has cost more or less than its budget
Schedule variance (SV)	$EV - PV$	+ ahead	- behind	Dollars' worth of work ahead of or behind plan
Cost performance index (CPI)	$EV \div AC$	> 1.00	< 1.00	Work you get for each dollar. 0.90 means 90 cents of work per dollar
Schedule performance index (SPI)	$EV \div PV$	> 1.00	< 1.00	Share of the planned work that got done
Variance at completion (VAC)	$BAC - EAC$	+ underrun	- overrun	How far over or under budget it will finish
Percent variances	$CV \div EV, SV \div PV$	Use these to compare work packages of different sizes		

3 Estimate at completion (EAC): four ways, one range

METHOD	FORMULA	ASSUMES	USE IT AS
CPI method	$BAC \div CPI$	Cost efficiency so far continues	Usually the floor. Overruns rarely finish below it
Composite	$AC + (BAC - EV) \div (CPI \times SPI)$	Being behind schedule will also cost money	Usually the ceiling
Budget rate	$AC + (BAC - EV)$	The rest of the work goes exactly to plan	Only when the cause was truly one-time
Bottom-up	$AC + \text{new ETC}$	A fresh estimate of the remaining work	Test it against the range above

Estimate to complete: $ETC = EAC - AC$. A contractor's own EAC is often called the LRE, latest revised estimate.

4 To-complete performance index (TCPI)

$$TCPI_{BAC} = (BAC - EV) \div (BAC - AC)$$

Efficiency the remaining work needs to finish **on budget**.

$$TCPI_{EAC} = (BAC - EV) \div (EAC - AC)$$

Efficiency needed to finish at the **current EAC**.

The reality check. Compare $TCPI_{BAC}$ with the CPI you have actually been getting. If TCPI is 0.05 to 0.10 or more above it, the plan is betting on a turnaround that almost never comes. Ask for a new EAC.

Reading EVM Like a Program Manager

What CPI and SPI are telling you

INDEX	SIGNAL	DO THIS
● 0.95 and up	Normal noise	Keep watching the trend
● 0.90 to 0.94	Drifting	Find the cause in the control accounts
● Below 0.90	Real trouble	Corrective action; revisit the EAC

A common convention, not a rule. Your contract's variance thresholds decide what must be explained.

The 20 percent rule. Once a contract is about 20 percent complete, cumulative CPI rarely improves by more than 10 percent, and it usually gets worse. An early overrun is a forecast, not a blip.

SPI fades. As work finishes, EV catches up to PV, so SPI drifts back to 1.00 even on a late program. Late in a contract, read the schedule's critical path, not SPI.

Where the data comes from: the IPMDAR

The Integrated Program Management Data and Analysis Report replaced the CPR (Formats 1 to 5) and the IPMR. It arrives as data, in three parts:

CPD Contract Performance Dataset: time-phased cost and hours

SPD Schedule Performance Dataset: the IMS

PNR Performance Narrative Report: variance explanations

DID DI-MGMT-81861, current revision D (March 2026). The old "Format 1" WBS view can still be produced from the CPD.

When DoD requires EVM

COST OR INCENTIVE CONTRACT VALUE	REQUIREMENT
\$50M and up	EVMS that complies with EIA-748 (was \$20M)
\$100M and up	DCMA-validated EVMS (was \$50M)

FY2025 NDAA §823, applied by DFARS class deviation 2026-O0011, effective Feb 1, 2026. Firm-fixed-price contracts are normally exempt. Contracts with EVM get an Integrated Baseline Review soon after award.

A different tripwire: Nunn-McCurdy

CPI measures one contract. Nunn-McCurdy measures a major program's **unit cost** against its Acquisition Program Baseline. **Significant breach:** 15 percent over the current baseline or 30 percent over the original. **Critical breach:** 25 percent over current or 50 percent over original, which forces a formal review and, unless the program is certified, termination. 10 U.S.C. §§4371-4377

Words you will hear in the review

PMB Performance measurement baseline: the time-phased budget EV is measured against

MR Management reserve: budget held back for in-scope surprises, outside the PMB

CA Control account: where scope, schedule and budget meet; owned by a CAM

IBR Integrated baseline review: joint check that the baseline is realistic

CBB Contract budget base: negotiated cost plus authorized, unpriced work

UB Undistributed budget: in the PMB but not yet assigned to a control account

WP Work package: the detailed task where EV is actually earned

OTB/OTS Over target baseline or schedule: a formal re-plan when the PMB is no longer achievable

Try it: a worked example

A \$10M contract. At month 12 the plan said \$4.0M of work (PV). The team finished \$3.6M of it (EV) and spent \$4.2M (AC).

$$CV = 3.6 - 4.2 = -\$0.6M$$

$$SV = 3.6 - 4.0 = -\$0.4M$$

$$CPI = 3.6 \div 4.2 = 0.86$$

$$SPI = 3.6 \div 4.0 = 0.90$$

$$EAC (CPI) = 10 \div 0.86 = \$11.7M$$

$$EAC (composite) = 4.2 + 6.4 \div (0.86 \times 0.90) = \$12.5M$$

$$TCPI = 6.4 \div 5.8 = 1.10$$

Read it: expect \$11.7M to \$12.5M, not \$10M. Finishing on budget would take 1.10 efficiency from a team running at 0.86, so plan for the overrun now.